

偏微分方程.

$$\begin{cases} \text{波动方程 } \partial_t^2 u = a^2 \Delta u + f \\ \text{扩散/热传导 } \partial_t u = a^2 \Delta u + f \\ \text{泊松 } \Delta u = f \end{cases}$$

$$a u_{xx} + 2b u_{xy} + c u_{yy} + H(x, y, u, u_x, u_y) = 0$$

$$\Rightarrow \text{特征方程 } a dx^2 + 2b dx dy + c dy^2 = 0$$

$$a(dy)^2 - 2b dx dy + c(dx)^2 = 0$$

↓

$$\Delta = b^2 - ac$$

$$\textcircled{1} \Delta > 0, \text{ 双曲型. } \frac{dy}{dx} = C_1, C_2 \text{ 积. 若是 } f(x),$$

$$\Rightarrow \text{令新变量 } \xi = y - C_1 x, \eta = y - C_2 x \text{ (特征线)}$$

$$\Rightarrow \text{代入: } u_x = u_\xi \cdot \xi_x + u_\eta \cdot \eta_x \dots \text{ 代为 } u_{\xi, \eta} \text{ 型.}$$

$$\text{令 } s = \frac{\xi + \eta}{2}, t = \frac{\xi - \eta}{2} \text{ 代入.}$$

$$\text{化简为 } u_{ss} - u_{tt} = \dots \text{ 型. 使 } u_{tt} = 0 \text{ 或类似.}$$

$$\textcircled{2} \Delta = 0, \text{ 抛物线型. } \frac{dy}{dx} = c.$$

$$\Rightarrow \xi = y - cx, \eta = x.$$

$$\Rightarrow u_{\eta\eta} / u_{tt} = \dots$$

齐次方程:

$$\text{要求 } \begin{cases} u_{tt} = a^2 u_{xx} \\ u|_{t=0} = 0 \\ u_t|_{t=0} = 0 \end{cases}$$

$$\begin{cases} w_{tt} = a^2 w_{xx} \\ w|_{t=0} = 0 \\ w_t|_{t=0} = f(x, t) \end{cases} \Rightarrow u = \int_0^t w(x, t, \tau) d\tau \text{ 而解.}$$

$$w(x, t, \tau) = \frac{1}{2a} \int_{x-a(t-\tau)}^{x+a(t-\tau)} f(z, \tau) dz.$$

分离变量法.

本征值: 使存在非零解而常数.

本征函数 (比如正弦, 余弦).

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$r = e^t, r' = \frac{1}{r} \frac{dr}{dt}, r'' = \frac{1}{r} \left(\frac{d^2 r}{dt^2} - \frac{dr}{dt} \right)$$

$$\textcircled{3} \Delta < 0 \text{ 椭圆型 } \frac{dy}{dx} = m + ni$$

$$\xi = \eta - mx, \eta = nx.$$

$$\Rightarrow u_{\xi\xi} + u_{\eta\eta} = \dots$$

特征法.

$$\textcircled{1} \begin{cases} u_{tt} = a^2 u_{xx}, t > 0. \\ u|_{t=0} = \varphi(x) \\ u_t|_{t=0} = \psi(x) \end{cases} \Rightarrow \text{达朗贝尔: } u(x, t) = \frac{1}{2} [\varphi(x-at) + \varphi(x+at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\alpha) d\alpha.$$

$$\textcircled{2} \begin{cases} u_{tt} = a^2 u_{xx}, x > 0, t > 0 \\ u|x=0 = 0, t > 0 \\ u|_{t=0} = \varphi(x), x > 0 \\ u_t|_{t=0} = \psi(x), x > 0 \end{cases}$$

奇延拓.

再取 $x > 0$ 部分

$$\varphi(x) = \begin{cases} \varphi(x), x > 0 \\ -\varphi(-x), x < 0 \end{cases}$$

代入要讨论.

$$\textcircled{3} \begin{cases} u_{tt} = a^2 u_{xx}, x > 0, t > 0 \\ u|x=0 = 0, t > 0 \\ u|_{t=0} = \varphi(x) \\ u_t|_{t=0} = \psi(x) \end{cases} x > 0.$$

偶延拓.

化简 $\Rightarrow$ ODE (积分) $\Rightarrow$ 边界条件.

非齐次: $u_{tt} = a^2 u_{xx} + f(x, t)$ 时, 令 $u = u_1 + u_2$.

使 u_1 齐次, u_2 无边界为 0.

解法

$$\textcircled{1} \text{ 代简: 只消二次: } u_{\xi\eta} = 0$$

u_{xy} 变量求特征线.

$$u_{xx} + 2c \cos y \cdot u_{xy} - \sin^2 x u_{yy} - \sin x u_y: t_2 = \frac{1}{4} u_{\xi\eta}.$$

$$\textcircled{2} \text{ 思路: ODE / 再求解 } F(\xi) + G(\eta) \text{ 一定.}$$

$$\textcircled{2} u_{tt} = a^2 u_{xx} + f(x, t).$$

$$\text{齐次: } u|_{t=0} = \varphi, u_t|_{t=0} = \psi, \text{ 则 } u = \frac{1}{2} [\varphi(x+at) + \varphi(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\alpha) d\alpha$$

$$\text{非齐: } \begin{cases} u_{tt} = a^2 u_{xx} \\ u|_{t=0} = \varphi \\ u_t|_{t=0} = \psi \end{cases} \text{ 与 } \begin{cases} u_{tt} = a^2 u_{xx} + f \\ \varphi \\ \psi \end{cases}$$

$$\begin{cases} w_{tt} = a^2 w_{xx} \\ \varphi \\ \psi \end{cases}$$

$$\Rightarrow \text{公式 } u = \int_0^t w d\tau$$

奇偶延拓: f, φ, ψ 都要

分离变量:

$$\textcircled{1} \text{ 求 } X: X(0) = X(l) = 0 \Rightarrow \lambda = \frac{n^2 \pi^2}{l^2}, n = 1, 2, X_n = \sin \frac{n\pi}{l} x.$$

$$X'(0) = X'(l) = 0 \Rightarrow \lambda = \frac{n^2 \pi^2}{l^2}, n = 0, 1, 2, X_n = \cos \frac{n\pi}{l} x$$

$$X'(0) = X(l) = 0 \Rightarrow \lambda = \frac{\pi^2}{l^2} (n + \frac{1}{2})^2, X_n = \cos (n + \frac{1}{2}) \frac{\pi}{l} x$$

$$X(0) = X'(l) = 0, \lambda = \frac{\pi^2}{l^2} (n + \frac{1}{2})^2, X_n = \sin (n + \frac{1}{2}) \frac{\pi}{l} x.$$

$$\textcircled{2} f(x) = \frac{a_0}{2} + \sum (a_n \cos n\theta + b_n \sin n\theta)$$

$$\frac{1}{2} \int_0^l f(x) dx \quad \frac{1}{2} \int_0^l f(x) \cos \frac{n\pi x}{l} dx.$$

$n \rightarrow \infty$

$$\textcircled{3} \text{ 圆环: } \Delta u = 0 \Rightarrow r^2 u_{rr} + r u_r + u_{\theta\theta} = 0 \Rightarrow R = A r^n + B r^{-n}$$

$$\text{若 } u(r, \theta) = u(r, \theta + 2\pi)$$

总结: 分离变量法

① 最基础: 齐次方程, 齐次边界条件

(1) $u_t = a^2 u_{xx}$, $u_{tt} = a^2 u_{xx}$, $u_{xx} + u_{yy} = 0$ 注意有没有0.
→ ① 设 $u = XT$; ② 找有齐次边界条件的方程, 列; ③ 得另一个.

④ 用没用过的条件, 代.

这里, (1) $\triangle$ 函数

(2) 傅立叶 $\rightarrow$ 分部积分.

about 傅立叶: $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\theta + b_n \sin n\theta)$

区间 $(0, l)$: $\int \frac{a_0}{2} = \frac{1}{l} \int_0^l f(x) dx$

$$\begin{cases} a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx \\ b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx \end{cases}$$

不管主方程有 $u_t/u \dots$ 只要有 $u(0,t)=u(\pi,t)=0$ 类似 π , 都直接展开.

→ 附加条件 $u(r, \theta) = u(r, \theta + 2\pi)$

(2) 圆环. $\Delta u = 0 \Rightarrow r^2 u_{rr} + r u_r + u_{\theta\theta} = 0$. (常 $n=0$ 也要) $\Rightarrow$ 欧拉. $\Rightarrow$ 令 $r = e^t$

② 若非齐次方程

$$\Rightarrow R = Ar^n + Br^{-n}$$

$f(x,t)$ 只含 t 或含 x/t : 参考 T10. 把 $f(x,t)$ 也分解成 $\sin$ 或 $\cos$.

依然设 $u = XT$, 得关于 T 的 ODE.

$f(x,t)$ 只含 x : 令 $u(x,t) = v(x,t) + w(x)$ 参考 T9.11.

凑出关于 v 的齐次方程. 齐次边界值 $\rightarrow$ 莫忘.

③ 若非齐次边界条件.

$\begin{cases} u|_{x=0} = p(t) \\ u|_{x=l} = q(t) \end{cases} \Rightarrow$ 构造 $u = v + w$, $w(x,t) = p(t) + \frac{x}{l} [q(t) - p(t)]$.
也可以让 w 同时满足 $\downarrow$

使 $\begin{cases} v|_{x=0} = 0 \\ v|_{x=l} = 0 \end{cases}$ 之后再回归正题求.

行波法:

① 二阶线性双曲型偏微分方程: 形如 $u_{xx} + 2u_{xy} - 3u_{yy} + \dots u_x = 0$

$\Rightarrow$ 特征线法. 特征方程 $\rightarrow$ 特征线 $\rightarrow$ 若只有二次项: $\Rightarrow u_{\xi\eta} = 0$

② $u_{tt} = a^2 u_{xx} + f(x,t)$ 型:

不然: 详细讨论 $\rightarrow$ 解 ODE / 特解

齐次: $u|_{t=0} = \varphi, u_t|_{t=0} = \psi(x)$ 设 $u(\xi, \eta) = F(\xi) + G(\eta)$

$$\Rightarrow u = \frac{1}{2} [\varphi(x-at) + \varphi(x+at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\alpha) d\alpha.$$

非齐: $\begin{cases} u_{tt} = a^2 u_{xx} \\ u|_{t=0} = \varphi \\ u_t|_{t=0} = \psi \end{cases}$ 与 $\begin{cases} u_{tt} = a^2 u_{xx} + f \\ u|_{t=0} = 0 \\ u_t|_{t=0} = 0 \end{cases}$

$\Downarrow$
公式

$$\begin{cases} w_{tt} = a^2 w_{xx} \\ w|_{t=0} = 0 \\ w_t|_{t=0} = f \end{cases} \Rightarrow \text{公式} \Rightarrow u = \int_0^t w d\tau$$

通常情形:
$$\begin{cases} X'' + \lambda X = 0 \\ X(0) = X(l) = 0 \end{cases}$$

需特别注意 $n=0$?

$$\Rightarrow \lambda = \frac{n^2 \pi^2}{l^2}, \quad n=1, 2, \dots \quad X_n = \sin \frac{n\pi}{l} x$$

凤凰朝圣 宝盖必胜!

$$\begin{cases} X'' + \lambda X = 0 \\ X'(0) = X(l) = 0 \end{cases}$$

$$\Rightarrow \lambda = \frac{\pi^2}{l^2} \left(n + \frac{1}{2}\right)^2, \quad X_n = \cos \left(n + \frac{1}{2}\right) \frac{\pi}{l} x$$

$$\begin{cases} X'' + \lambda X = 0 \\ X(0) = X'(l) = 0 \end{cases}$$

$$\Rightarrow X_n = \sin \left(n + \frac{1}{2}\right) \frac{\pi}{l} x$$

$$\lambda = \frac{\pi^2}{l^2} \left(n + \frac{1}{2}\right)^2$$

对 T : $T_n = K_n e^{-\lambda_n a^2 t}$

$$\begin{cases} X'' + \lambda X = 0 \\ X'(0) = X'(l) = 0 \end{cases}$$

$$\Rightarrow \lambda = \frac{n^2 \pi^2}{l^2}, \quad n=0, 1, \dots \quad X_n = \cos \frac{n\pi}{l} x.$$

傅立叶变换

积分变换法

$$\hat{f}(\lambda) = \int_{\mathbb{R}} f(x) e^{-\lambda \cdot x i} dx$$

$$f^v(x) = \frac{1}{2\pi} \int_{\mathbb{R}} f(\lambda) e^{i\lambda x} d\lambda.$$

$$\Rightarrow \text{即 } f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \left[\int_{\mathbb{R}} f(x) e^{-\lambda x i} dx \right] e^{i\lambda x} d\lambda$$

性质: 线性; ...

$$\text{微分: } [f^{(m)}]^\wedge(\lambda) = (i\lambda)^m \hat{f}(\lambda)$$

$$F[f(x)] = f(x) \rightarrow F(\lambda) \quad F[f(x)] = F(\hat{x})$$

① 线性: $F(af(x) + bg(x)) = aF(f(x)) + bF(g(x))$.

② 平移: $F(f(x-h)) = e^{-ih\lambda} F(\lambda)$

$$f(x)e^{ihx} \rightarrow F(\lambda-h)$$

③ 伸缩性: $f(ax) \rightarrow \frac{1}{|a|} F(\frac{\lambda}{a})$

④ 乘子: $xf(x) \rightarrow i \frac{d}{d\lambda} F(\lambda)$ ✓

⑤ 对称 $F(x) \rightarrow 2\pi f(-\lambda)$

⑥ 伴随 $\int f(x)g(x)dx = \int F(\lambda)g(\lambda)d\lambda$ ✓

⑦ 微分 $f^{(n)}(x) \rightarrow (i\lambda)^n F(\lambda)$ ✓

⑧ 积分: $\int_{-\infty}^{\infty} f(t)dt \rightarrow \frac{1}{i\lambda} F(\lambda)$

⑨ 守恒: $\int_{-\infty}^{\infty} f(x)^2 dx = \frac{1}{2\pi} \int_{-\infty}^{\infty} [F(\lambda)]^2 d\lambda$

⑩ 卷积: $F[f * g] = F[f]F[g]$ ✓

$$e^{-x^2} \rightarrow \sqrt{\pi} e^{-\frac{1}{4}\lambda^2}$$

$$e^{-ax^2} \rightarrow \sqrt{\frac{\pi}{a}} e^{-\frac{\lambda^2}{4a}}$$

高斯积分.

并时常用到: $\int_{-\infty}^{\infty} e^{-k^2} dk = \sqrt{\pi}$

$$\frac{1}{\sqrt{2\pi}} F[f(\lambda)]$$